

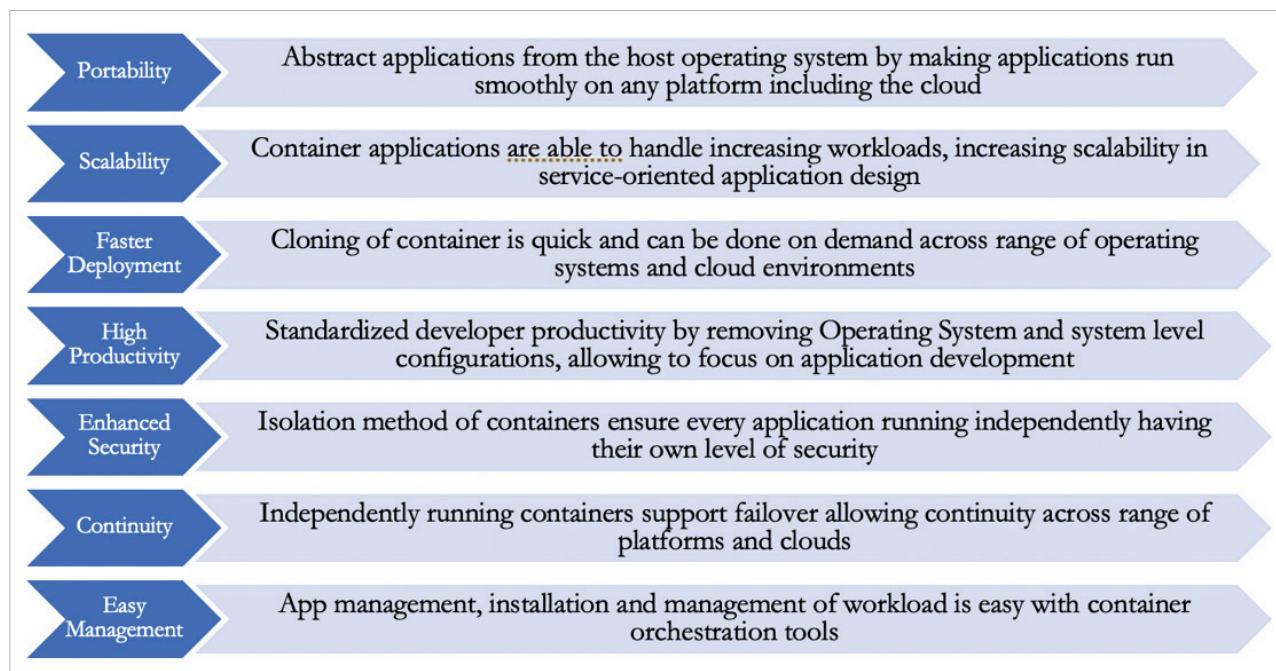


Figure 1: Benefits of containerization

Apache Mesos and OKD. Though there are many commercial container solutions, open source solutions are not only free but have rich features and a wide developer adoption base. Let's look at a few of them briefly, and the unique capabilities and functionality they offer.

Kubernetes is a Linux based open source container solution for cluster computing. The ease of managing organised applications is a key allure for developers and DevOps engineers. Creating an app, and specifying a pipeline and type of workload is what it takes for Kubernetes to automatically run in an environment. This feature helps Kubernetes score over traditional container-based applications with respect to agility and reliability. Notification is built in for monitoring application deployment and execution statistics, making it easy for developers to focus on scale and performance. Once scale and security are addressed to the needs of the organisation, Kubernetes runs on autopilot for container-based applications to grow into whatever environment organisations choose, including the cloud.

Docker is yet another popular and widely adopted containerization technology, which works well on the Kubernetes platform. The speed at which containers can be created is one of the key benefits of Docker.

Apache Mesos is an open source platform for automating deployment, scaling, and management of large clusters of container based applications. Ease of customisation for a wide range of applications is a significant differentiator of Mesos. The ability to run Dockers and other such programs across the distributed network is one of the reasons many organisations adopt Mesos.

OKD is an open source alternative to OpenShift with similar features and functionality. One of its top benefits is the ability to automatically distribute Kubernetes on all popular cloud platforms (AWS, Azure, GCP) and bare metal, OpenStack, Red Hat, and other virtualization providers. Quick application build, scaling, support for automatic high availability, load balancing, health check, and failover are the features that attract organisations to OKD.

In addition to these tools, there is a significant amount of innovation in containerization simplification in terms of maintenance and providing an additional layer of robustness. A good example is Toradex, which offers rich features when it comes to seamless integration with hardware, operating system, development tools, and remote management including updates and monitoring. The platform is highly reliable and secure. Toradex looks to increase efficiency across the board in the software development life cycle by providing easy-to-maintain products, and integration over-the-air including updates and device monitoring. This allows development teams to focus on what's important to the business. Its operating system Torizon takes advantage of the large open source ecosystem to integrate Docker Hub and thousands of ready-to-use containers. This open source platform is customisable, and can be deployed on different hardware. And it's built on open standards with open APIs to make integration with other open source and third party services